

ON DEVICE · everything safety-critical runs here

SENSE

ARKit pose

LiDAR depth 256 × 192

Camera · Core ML

Microphone · on-device STT

DECIDE

Circuit counter

Path guide

Obstacle ranker

Voice intent · fixed grammar

SPEAK

Spoken guidance · EN/AR/UR

Spatial audio beacon

Haptics

▲ safety warnings interrupt all other audio

NETWORK · optional — every path has an on-device fallback

Scene description
1 frame, on request

Relocalisation
Kaaba as anchor

SOS +
location share

Companion
booking

FAMILY PORTAL

Method

The Kaaba was simulated as an ARKit world anchor placed on a physical object, and testers walked a 2 m proxy course indoors. Approximately 20 trials across friends and co-workers on iPhone 16 Pro, plus 44 headless assertions across four ritual-logic simulations. Obstacle detection runs YOLOv8n (80 COCO classes) converted to Core ML, with Apple's Vision human detector as fallback; each detection is ranged by the LiDAR depth map.

Verified on device

Tawaf circuit counting



Circle guidance — drift inward / outward



Obstacle detection + LiDAR range



Voice command layer — EN / AR / UR



Sa'i mode and guidance



Du'a with safety interrupt



Web portal · caregiver matching · SOS — pilot pending: needs a larger cohort and resources beyond the hackathon

Still to run: a pilot with pilgrims who have visual impairment.